

Improving the Security of Georgia Tech's Open-Source Ecosystem at Scale

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Georgia Tech researchers, labs, and students maintain thousands of public open source repositories. Across these repositories there are common gaps in best practice security measures; many of these security gaps are simple but impactful to address.

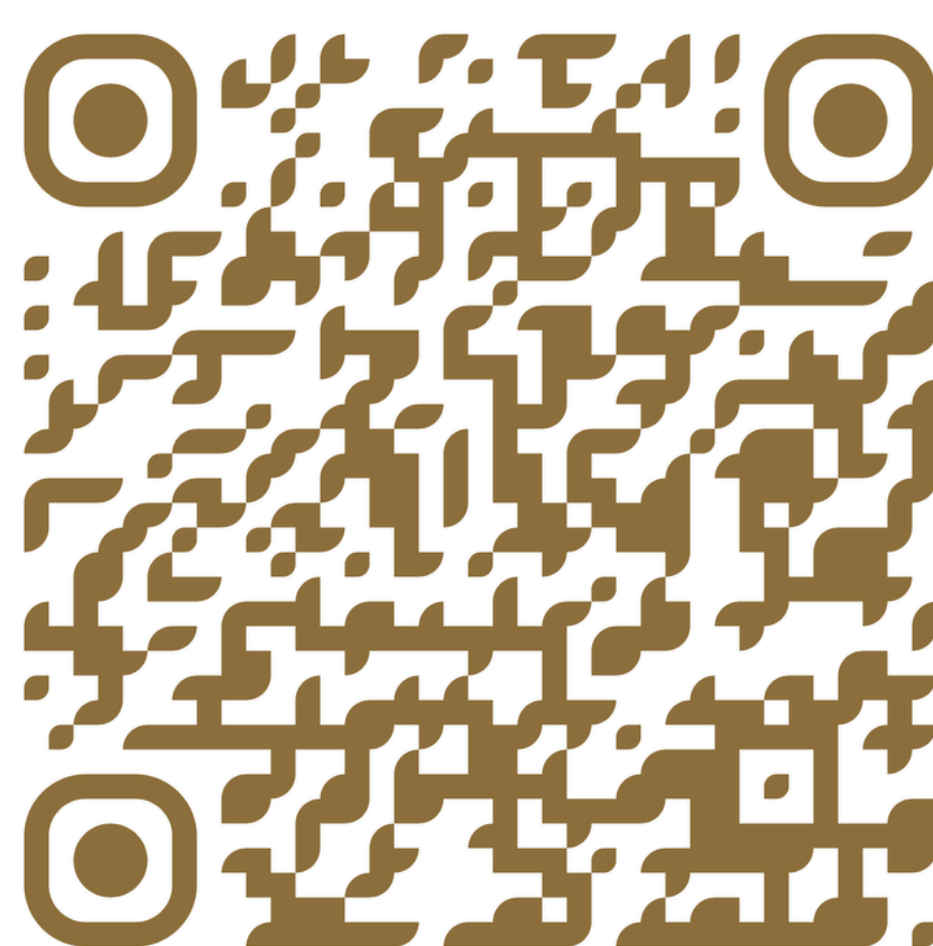
The goal is to evaluate widely used Georgia Tech open-source repositories and develop automated tooling to improve their security posture at scale.

1

Find impactful GT open source repos

- Scrape GitHub for repositories mentioning Georgia Tech, then confirm affiliation with an LLM classifier.
- Merge in community engagement metrics, stars, forks, watchers, last maintained to rank by real-world impact and activity

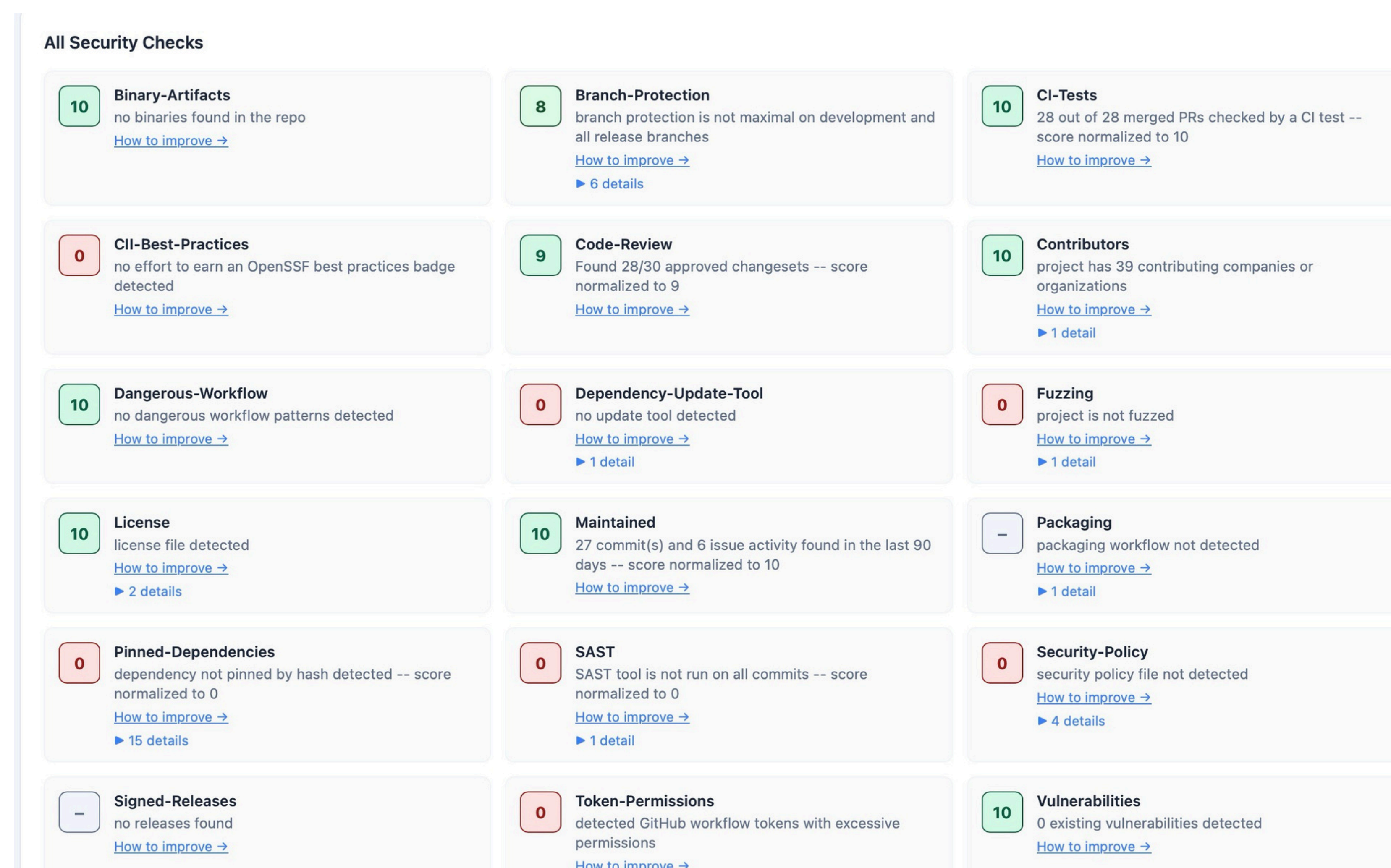
Learn More:



2

Analyze missing security best practices

- Automate running Open Source Security Foundation (OpenSSF) scorecards across GT repositories to provide a score on known vulnerabilities, missing branch protection, unpinned dependencies, absent code review, and weak CI/CD practices, and more.
- Visualize and identify high-priority security gaps



3

Send Automated PR's and Reports

- For each recurring gap, a script forks the repo, generates the missing file, and opens a pull request upstream. (SECURITY.md, dependabot.yml, and LICENSE built so far, with more to come)
- Each PR explains what the file does and why it matters, so maintainers can review in one sitting.

